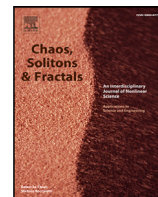




Contents lists available at ScienceDirect

Chaos, Solitons and Fractals

journal homepage: www.elsevier.com/locate/chaos

An efficient logic-guided attack strategy on logical-physical coupled networks

Mengying Geng^a, Ling Zhang^a, Min Hui^a, Yiming Huang^b, Feng Qing^a,
Mengqiao Xu^c, Changjun Fan^{a,*}, Guangquan Cheng^a^a Big Data and Decision Laboratory, National University of Defense Technology, Hunan 410000, China^b Department of Computing, Imperial College London, London SW7 2AZ, UK^c Dalian University of Technology, Dalian 116000, China

ARTICLE INFO

Keywords:

Complex networks
Centrality measure
Network disruption
Multi-layer coupled networks
Cross-layer dependency analysis

ABSTRACT

In an era of increasingly interconnected and complex systems, conventional network attack strategies often prove inadequate when confronting the structural and functional intricacies of multi-layer infrastructures. Logical components often lack tangible substances and thus cannot be targeted directly by physical means. Moreover, failures may cascade across layers due to the intricate interweaving of logical and physical layers, further limiting the effectiveness of conventional attack approaches in multi-layer scenarios. To address these challenges, we propose the Critical Edge-to-Node Strike (CENS) framework, which leverages logical-layer guidance to induce efficient damage at the physical layer. CENS bridges logical-physical dependencies by assessing the hybrid centrality of logical edges and employing a max-flow-based graph-cut algorithm to map logical connection damage onto physical nodes. We further introduce a node ablation mechanism that transforms the physical topology to efficiently identify fundamental logical dependencies. This mechanism simplifies path enumeration, improves stability, and enhances attack precision. Extensive experiments on both IEEE118 system and synthetic networks reveal that CENS and its ablation-enhanced variant outperform centrality-based baselines in terms of logical damage ratio, speed, and robustness, achieving up to 60% logical failure with only 15% physical nodes removal. These results demonstrate the effectiveness and superiority of the strategies in logical-physical coupled network attack scenarios.

1. Introduction

Modern infrastructure systems and information networks are increasingly interconnected and multi-layered, integrating physical components with overlaying logical control or communication layers. Complex network models have long been used to represent such systems, abstracting components as nodes and interactions as edges. Aided by topological analysis, statistical methodology (e.g., maximum entropy models), natural language processing techniques, etc., this paradigm has been proven valuable for analyzing empirical systems ranging from power grids to communication and social networks [1]. In parallel, there is a growing need for effective strategies to disrupt or protect these networks, often termed attack strategies in network science [2,3], motivated by applications in domains such as military operations, information dissemination, and epidemic containment. The development of

* Corresponding author.

E-mail addresses: gengmengying@nudt.edu.cn (M. Geng), zhangling@nudt.edu.cn (L. Zhang), minhui@nudt.edu.cn (M. Hui), yiming_huang@std.uestc.edu.cn (Y. Huang), qingfeng@nudt.edu.cn (F. Qing), stephanie@dlut.edu.cn (M. Xu), fanchangjun@nudt.edu.cn (C. Fan), cgq299@nudt.edu.cn (G. Cheng).

<https://doi.org/10.1016/j.chaos.2025.117396>

Received 15 July 2025; Received in revised form 31 August 2025; Accepted 6 October 2025

Available online 23 October 2025

0960-0779/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

robust attack and defense strategies in networked systems is considered essential for revealing system vulnerabilities and facilitating the design of resilient infrastructures.

Recent studies have shown that the interdependence across multiple network layers significantly amplifies vulnerabilities beyond what is observed in single-layer settings. Classic works such as Buldyrev et al. [4] and Gao et al. [5] demonstrated that bidirectional dependencies or even small degrees of interdependence can trigger cascading failures and abrupt phase transitions under targeted attacks. [6,7] further highlighted how spatial coupling distance and symmetry in cross-layer dependencies affect robustness. These results suggest that multilayer architecture critically governs failure propagation, underscoring the necessity of attack strategies tailored for coupled networks.

Building on these insights, various attack strategies have been proposed for multilayer or interdependent networks. For example, Zhang et al. [8] introduced a bidirectional coupling-based optimization approach (BICND-EMT) to balance efficiency and cost; Singh et al. [9] employed graph coloring to mitigate cascading failures by shielding medium-degree nodes; and Ribeiro et al. [10] applied graph representation learning (GraphSAGE) to identify key actors across heterogeneous layers.

While the aforementioned works have significantly advanced the design of efficient network attack strategies, several critical practical constraints remain. In particular, nodes in the logical layer (e.g., communication units) are abstractly represented entities and therefore cannot be directly targeted. Instead, they can only be disrupted indirectly by launching attacks against their corresponding counterparts in the physical layer. Furthermore, the inherent coupling between the physical and logical layers is known to pose significant and often intractable challenges for the design of effective attack strategies.

Motivated by these challenges, this study introduces a refined multilayer network model explicitly capturing physical–logical coupling characteristics in coupled systems. At the theoretical level, we rigorously prove through detailed mathematical derivation that the overall functionality of the system depends on a subset of logical edges with specific structural configurations, which are mapped onto physical-layer nodes and are explicitly identified during the analytical process. Based on this insight, we propose the Critical Edge-to-Node Strike (CENS) strategy, a novel coordinated attack approach that efficiently targets critical physical-layer nodes guided by logical connectivity. To further optimize the performance of CENS, we introduce a node ablation mechanism that reduces the number of targeted physical nodes while preserving or enhancing overall attack effectiveness. Extensive experiments conducted on empirical and synthetic datasets validate that CENS achieves superior damage rates, accelerated attack speeds, and improved generalizability. Specifically, our proposed approach demonstrates that removing merely 15% of physical nodes effectively disrupts 60% of logical connectivity, with damage propagation speeds 40% faster than baseline methods, underscoring the strategic advantage of exploiting physical–logical coupling.

The main contributions of this paper are summarized as follows:

- Developing an enhanced multilayer network model explicitly accounting for physical–logical layer coupling, theoretically elucidating how specific logical-to-physical mappings underpin system functionality, and rigorously proving that system functionality depends on a subset of logical edges with specific structural configurations through detailed mathematical derivation.
- Based on theoretical findings, we propose the Critical Edge-to-Node Strike (CENS) strategy, enabling efficient identification of physical-layer targets guided by logical connectivity.
- We further introduce a Node Ablation Mechanism that enhances CENS by reducing the number of target nodes without sacrificing attack performance.
- Extensive experiments on both empirical and synthetic datasets show that CENS achieves higher damage rates, faster attack speed, and better generalization.

The rest of the paper is organized as follows: Section 2 reviews related work. Section 3 formalizes the physical–logical coupled network architecture and analyzes cross-layer failure mechanisms. Section 4 introduces the proposed coupled network attack method, detailing both the CENS strategy and its node-ablation extension. Section 5 presents a comprehensive comparison of the proposed methods against baseline methods on both empirical and synthetic networks. Finally, we discuss and conclude in the last section.

2. Related works

Over the past decades, numerous studies have explored how to identify the most critical nodes whose removal maximizes network damage [3,11]. Approaches can be broadly categorized into three categories: centrality-based methods, heuristic strategies, and machine learning-based approaches. Early research focused on using topological metrics to identify critical nodes, introducing classical centrality measures such as Degree Centrality (DC) [12], Betweenness Centrality (BC) [13], and Closeness Centrality (CC) [14]. DC ranks nodes by their number of direct connections, offering simplicity but limited global insight. BC evaluates a node's control over information flow by counting shortest paths passing through it, but it is computationally expensive and may overlook locally influential nodes. To reduce complexity, Bartolucci et al. proposed a rank-1 approximation approach that estimates node influence from only local aggregate information – without full topology or matrix inversion – making key node identification feasible in large-scale or incomplete networks [15]. CC measures how close a node is to all others via shortest paths, capturing global influence yet being sensitive to disconnected components. However, classical metrics often fail to capture the nuanced role of nodes in large or complex networks due to either locality or oversimplified assumptions [16–19]. For example, traditional centralities fail to distinguish centrality distribution differences across nodes with varying degrees. By contrast, cavity method-based centrality analysis derives recursive Katz centrality distribution equations for undirected random networks [20]. It uncovers centrality's multi-modal

distribution, provides a “null model” benchmark for empirical network key node identification, and helps distinguish “over-expressed” vs. “under-expressed” node centrality, addressing traditional methods’ limitations in characterizing global distributions. Beyond this research approach, there exist other thoughts aimed at addressing the limitations of classical centrality metrics. For instance, some studies focus on integrating localized and global structural features. Local Information Dimensionality (LID) dynamically adjusts box sizes to optimize critical node identification [21], while Zhang et al. incorporated Laplacian centrality into gravity models for enhanced robustness [22]. The Local Degree Dimension (LDD) algorithm further refines identification by modeling neighbor growth or decay to reflect node dimensionality [23]. To reduce computational complexity, Zhao et al. extended the gravity model by using walk distances instead of shortest paths, making it scalable to large networks [24]. Additionally, Vertex Entanglement (VE) evaluates the effect of local perturbations on global topology to detect influential nodes [25]. DomiRank integrates both local and global topological information for efficient and accurate identification of vulnerable nodes [26]. Although centrality-based attacks are physically interpretable, they may overlook vulnerabilities arising from functional dependencies or application-specific dynamics. In response, various heuristic algorithms are devised to improve attack efficiency. A notable example is Arulselvan’s greedy algorithm for node removal via maximal independent sets [27], followed by Aringhieri’s Variable Neighborhood Search (VNS), which reduces time complexity [28]. Shen et al. address heterogeneous cost constraints in hub disruption using a simulated annealing approach [29]. In parallel, the rapid development of machine learning has opened up new avenues for developing adaptive and data-driven attack strategies. A representative example is FINDER [30], which enables cross-scale strategy transfer through deep reinforcement learning. The Smart-Core model [31] and vulnerability mapping approach [32] use machine learning techniques in real-world infrastructure. ISMnet [33] leverages higher-order convolution networks [34] to mine influential nodes and higher-order structures. Despite these advances, most existing methods implicitly assume a single-layer network structure and do not fully address the challenges posed by systems with multiple interdependent layers.

To capture the multifaceted interactions that underpin empirical systems, complex multilayer networks have gained increasing attention. Multilayer networks comprise interconnected layers, with each layer corresponding to different functions or types of interactions, such as social networks with relationship-specific layers [35] or transportation systems integrating subway and bus networks [36]. One important subclass consists of interdependent (or coupled) networks, in which the functionality of nodes in one layer depends on the nodes in another layer [37]. Such dependency exists across a wide range of scenarios, including economic input–output networks and ecological food webs [15,38,39]. The key attributes of these systems can all be accurately estimated using local aggregate information, providing cross-domain methodological references for identifying key nodes in the functional layers of multilayer coupled networks. In fact, the functional dependencies that shape economic or ecological systems manifest even more prominently in infrastructure-related multilayer coupled networks—where their presence not only defines how layers operate in tandem but also introduces unique systemic risks that demand targeted analysis. A representative example is the Power-Communication coupled network, comprising a physical layer (e.g., the power grid and the electricity factories) and a logical layer (e.g., a communication network). The communication layer depends on the grid’s functionality, while communication nodes simultaneously control and monitor the power infrastructure [40]. This mutual dependency introduces potential risks known as cascading effects, i.e., failures in one layer can propagate and induce damage in another layer. Notably, the cascading effect is the characteristic feature of coupled networks, where the functional interdependence between layers creates potential points of systemic vulnerability.

Researchers in network science have extensively studied cascading failures and network layer dependencies on network structure [41–45]. The manner in which the removal of critical edges leads to overloads in scale-free networks was examined by Holme et al. [46]. Early works primarily address single-layer network scenarios, demonstrating how targeted node or edge removals could trigger significant effects through mechanisms like load redistribution [46,47] and optimal path disruptions [48]. To more accurately characterize network dynamics, spatiotemporal and nonlinear processes have also been incorporated into the modeling framework. In this context, Bak’s sandpile model has been used to highlight system criticality through power-law failure distributions [49], while the Coupled Map Lattices (CML) model has been employed to explore spatial propagation phenomena via coupled map structures [50]. These studies reveal how topological structures correlate with failure propagation potential. Moreover, spatiotemporal and nonlinear dynamics are further introduced to provide a more accurate network description [51–53].

Moving beyond single-layer frameworks, recent research increasingly explores multilayer network settings, highlighting how cross-layer dependencies significantly amplify network vulnerabilities. Buldyrev et al. [4] leverage percolation theory to demonstrate how bidirectional dependencies amplify cascading effects in interdependent networks. It has been shown by Gao et al. [5] that low levels of interdependency can trigger abrupt, first-order phase transitions under targeted attacks, thereby challenging the conventional assumption of continuous failure dynamics. The impact of spatial coupling distance on systemic collapse thresholds has been rigorously investigated by Dong et al. [6], who demonstrate that longer coupling distances raise the threshold for collapse. Liu et al. [7] observe that a symmetric dependency across layers could sometimes enhance robustness by dampening feedback loops. These insights underscore that multilayer architecture – not merely intra-layer topology – governs failure propagation. Based on these insightful efforts, more recent approaches have optimized attack strategies under coupling constraints. Zhang et al. [8] introduce a Bidirectional Inter-layer Coupling Network Disruption and Evolutionary Multi-Task Optimization (BICND-EMT) to balance attack efficiency and cost. Singh et al. [9] leverage graph coloring to reduce failure propagation by shielding medium-degree nodes. Ribeiro et al. [10] apply the GraphSAGE model to detect key actors in criminal networks by aggregating relational cues across heterogeneous layers. For cross-layer failure analysis, Mean First Passage Time (MFPT) estimates failure propagation speed based on local transition probabilities of target nodes [39].

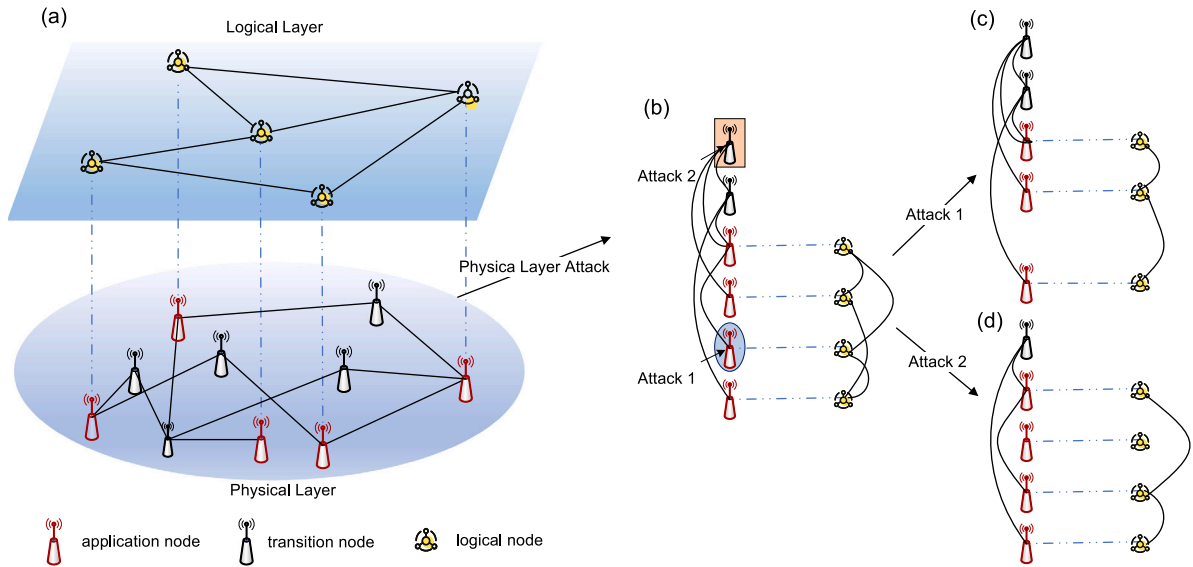


Fig. 1. Schematic of the coupled physical-logical network structure. The physical layer contains both application nodes (supporting functional services) and transition nodes (relaying communication), while the logical layer abstracts application-level interactions. Panels (b)–(c) show that removing an application node eliminates its corresponding logical node and all adjacent logical edges, causing cross-layer failure. Panels (b)–(d) illustrate that removing a transition node breaks connected physical links, disrupting both direct and multi-hop logical paths reliant on them.

3. Physical-logical coupled network

In this section, we first define the two-tier coupled network architecture, then analyze how removing application and transition nodes affects both physical and logical layers, and finally provide a rigorous theoretical analysis of cross-layer path invalidation.

3.1. Coupled network modeling

This paper proposes a two-tier coupled network architecture, consisting of a physical layer that encodes the spatial distribution and connectivity of physical entities, and a logical layer that captures their functional interdependencies. Fig. 1(a) illustrates the schematic structure of this coupled network. These two layers are tightly coupled: while logical interactions are abstracted from the physical infrastructure, their feasibility and resilience are constrained by the underlying physical connectivity. In our subsequent analysis, all attacks are performed solely on the physical layer; however, the resulting failures are reflected in the logical layer as well. As shown in Fig. 1, only nodes in the physical layer are directly attacked, but the impact propagates to the logical layer through interdependencies.

Formally, the physical layer is modeled as an undirected graph $G_p = (V_p, E_p)$, where V_p denotes the set of all physical nodes and E_p denotes the set of physical links. Based on their functions, nodes in V_p are categorized into two disjoint subsets: application nodes V_A and transition nodes V_T , such that $V_p = V_A \cup V_T$ and $V_A \cap V_T = \emptyset$. Application nodes represent function-critical entities tailored for specific services, and may vary across different scenarios. In contrast, transition nodes serve as intermediaries to relay information among application nodes. Each node $v_p^i \in V_p$ is assigned a binary type indicator z_p^i as:

$$z_p^i = \begin{cases} 1, & \text{if } v_p^i \in V_A \\ 0, & \text{if } v_p^i \in V_T. \end{cases} \quad (1)$$

The logical layer is modeled as another undirected graph $G_l = (V_l, E_l)$, where the logical node set V_l corresponds one-to-one with the application node set V_A , and the logical edge set E_l captures the abstracted functional interactions among logical nodes. Importantly, logical connectivity is physically constrained: a logical edge $\langle v_l^i, v_l^j \rangle \in E_l$ is valid only if a path exists in the physical layer between the corresponding application nodes. Therefore, logical communication is inherently dependent on physical-layer feasibility.

This architecture embodies an edge-oriented one-to-multiple mapping mechanism: each logical edge corresponds to one or more feasible paths composed of physical-layer links, while each physical link may concurrently support multiple logical interactions. It is assumed that all logical nodes have corresponding physical counterparts and that each logically connected node pair must be physically reachable through at least one path.

For the proposed physical-logical coupled network architecture, it is challenging to directly eliminate logical nodes, since the logical nodes is in essence an abstract description from the functional perspective. Therefore, this paper further investigates the relationship between the physical and logical layers and designs efficient attack strategies to remove physical layer nodes.

3.2. Physical layer attack analysis

In the following, we first illustrate the effect of removing application/transition nodes in both physical and logical layers, and then analyze their coupling relationship.

The physical layer consists of application nodes and transition nodes, where the application nodes form the logical layer and the transition nodes transfer information between application nodes. The removal of the application node from the physical layer firstly leads to the systematic disconnection of all adjacent physical links, causing intra-layer connectivity loss. Notably, the failure mechanism demonstrates through cross-layer functional dependencies: when the destroyed physical entities maintain direct service-provisioning relationships with logical nodes, the corresponding logical nodes and their interconnection links undergo synchronized elimination, as shown in Fig. 1(b)–(c).

The mechanism of removing transition nodes in the physical layer, which is shown in Fig. 1(b)–(d), is the same as the removal of application nodes. The elimination process involves not only the target transition node itself but also all its associated intra-layer connections. This deletion operation alters the topological structure of the physical layer, directly affecting the functions that rely on these connections for information transfer.

The impact of physical-layer transition node removal on logical connectivity necessitates targeted edge elimination in the logical layer. In the logical layer, the affected node pairs D_l exhibit two connectivity configurations: the direct connection and multi-hop pathways. For application node pairs D_l experiencing physical-layer path disruption, their logical-layer counterparts D_l undergo systematic disconnection through two processes: Direct edge deletion and Indirect path invalidation.

Direct edge deletion removes all adjacent edges connecting logically co-located node pairs:

$$E'_l = E_l \setminus \{ \langle v^i, v^j \rangle \mid (v^i, v^j) \in D_l \}.$$

Indirect path invalidation, however, experiences more complex steps. The path dependency indicates that, for each node pair (u, v) in D_l that was linked through multi-hop paths set $P(u, v)$, there would be at least one corresponding path in the physical layer between their associated physical-layer nodes. Specifically, if such logical paths exist: If $P(u, v) \neq \emptyset$, then there must exist at least one physical-layer path, i.e., $\exists p_p^i : a_u \rightarrow v_p^1 \rightarrow \dots \rightarrow v_p^k \rightarrow a_v$, where a_u and a_v are the physical nodes corresponding to u and v via the inverse mapping:

$$a_u = \psi^{-1}(u), \quad a_v = \psi^{-1}(v).$$

The full set of physical paths is then defined as $P_p(a_u, a_v) = \bigcup_{i=1}^n p_p^i$, $|P_p(a_u, a_v)| \geq 1$.

For the indirect path invalidation, let $D = \{ \langle \psi(a_i), \psi(a_j) \rangle \in E_l \mid (\psi(a_i), \psi(a_j)) \in D_l \}$ be the deleted edge set after the direct edge deletion. Crucially, the proposition below establishes that the removal of adjacency edges in the preceding step inherently guarantees the disruption of all indirect paths:

$$\forall p \in P(\psi(a_u), \psi(a_v)), \exists \langle \psi(a_i), \psi(a_j) \rangle \in D \cap E(p).$$

We will rigorously prove this proposition in Section 3.3.

3.3. Theoretical analysis

This section provides a rigorous graph-theoretical justification for how the removal of critical nodes in the physical layer ensures the complete disruption of all communication paths between corresponding node pairs in the logical layer. The results establish essential theoretical foundations for the subsequent development of targeted attack strategies.

Proposition 1 (Path Invalid Guarantee). Let $G_l = (V_l, E_l)$ denote the logical graph, and $D_l \subseteq V_l \times V_l$ be the set of logical node pairs whose direct adjacency is removed due to dependency disruption in the physical network $G_p = (V_p, E_p)$. Let $\psi : V_l \rightarrow V_p$ be the node-to-application mapping, and let $V_A = \text{Im}(\psi) \subseteq V_p$ denote the set of application nodes. After direct logical edge removal, the residual logical graph is denoted as $G'_l = (V_l, E'_l)$ with:

$$E'_l = E_l \setminus \{ \langle u, v \rangle \mid (u, v) \in D_l \}. \quad (2)$$

Then, for any node pair $(u, v) \in D_l$ that are not directly connected in G'_l , and any remaining multi-hop path $p \in P(u, v)$ in G'_l , it holds that:

$$\forall p \in P(u, v), \quad E(p) \cap D_l \neq \emptyset \Rightarrow \neg \text{path}(u, v \mid G'_l). \quad (3)$$

In the following, we provide a rigorous, constructive multi-step argument demonstrating that each residual logical path between $(u, v) \in D_l$ inevitably relies on at least one direct logical edge that has already been invalidated.

Proof. Step 1: Physical mapping and multi-hop assumption.

Consider a logical node pair $(u, v) \in D_l$ connected only via multi-hop logical paths

Let $P(u, v)$ denote all possible simple paths between u and v in G'_l . For any $p \in P(u, v)$, denote its structure explicitly as:

$$p = u \rightarrow v^1 \rightarrow \dots \rightarrow v^n \rightarrow v, \quad V(p) = \{u, v^1, \dots, v^n, v\}.$$

Each logical node is mapped via ψ to its application node in V_A :

$$\forall x \in V(p), \quad \psi(x) \in V_A.$$

Let $\psi(u) = a_u$, $\psi(v) = a_v$; we consider the set of all physical paths between (a_u, a_v) , denoted:

$$P_p(a_u, a_v) := \{\text{All physical paths from } a_u \text{ to } a_v \text{ in } G_p\}.$$

Step 2: Multi-hop guarantee in physical paths.

Suppose there exists a path $p_p^i \in P_p(a_u, a_v)$ such that:

$$|p_p^i| = 1 \Rightarrow \langle a_u, a_v \rangle \in E_p.$$

However, by assumption $\langle u, v \rangle \notin E_l$ after initial removal, which implies:

$$\langle a_u, a_v \rangle \notin E_p \Rightarrow \forall p_p^i, \quad |p_p^i| \geq 2.$$

Hence, all paths in $P_p(a_u, a_v)$ must include at least one intermediate physical node.

Step 3: Existence of intermediate application nodes.

Consider a physical path:

$$p_p^i = a_u \rightarrow v_p^1 \rightarrow \dots \rightarrow v_p^k \rightarrow a_v, \quad V(p_p^i) = \{a_u, v_p^1, \dots, v_p^k, a_v\}.$$

Assume for contradiction that:

$$\forall v_p^m \in V(p_p^i), \quad v_p^m \notin V_A \Rightarrow V_A \cap V(p_p^i) = \emptyset.$$

Then the logical path p corresponding to p_p^i contains no intermediate logical nodes, implying:

$$V(p) = \{u, v\} \Rightarrow \langle u, v \rangle \in E_l,$$

which contradicts the initial condition $\langle u, v \rangle \in D_l$. Hence:

$$\exists v_p^m \in V(p_p^i), \quad v_p^m \in V_A,$$

i.e., every physical path between a_u and a_v includes at least one intermediate application node.

Step 4: Recursive subpath dependency.

Each p_p^i maps to a logical path p with at least one intermediate logical node m , such that:

$$p = (u \rightsquigarrow m) \cup (m \rightsquigarrow v),$$

where each sub-paths are defined as:

$$p_p^{i1} = a_u \rightsquigarrow a_m, \quad p_p^{i2} = a_m \rightsquigarrow a_v.$$

Then, $p_p^i = p_p^{i1} \cup p_p^{i2}$ and both p_p^{i1}, p_p^{i2} are physical paths corresponding to shorter logical sub-paths.

If p_p^{i1} is disrupted due to transition node removal, then:

$$(u, m) \in D_l \text{ and } |p(u, m)| < |p(u, v)|.$$

This recursive reduction proceeds until the path reduces to:

$$\lim_{k \rightarrow 1} \min_p |p| = 1 \Rightarrow \{\langle x, y \rangle \in E_l \mid (x, y) \in D_l\}.$$

Hence, every sub-path in the original p contains at least one edge in D_l .

Consequently, every path $p \in P(u, v)$ satisfies:

$$E(p) \cap D_l \neq \emptyset \Rightarrow \neg \text{path}(u, v \mid G'_l). \quad \square$$

This proposition provides a sufficient condition for the complete disruption of logical-layer connectivity. By strategically invalidating specific logical edges through targeted physical node removal, it is possible to effectively disable all multi-hop logical paths. This theoretical foundation informs the subsequent Critical Edge-to-Node Strike (CENS) strategy by clarifying the critical dependencies between physical and logical layers and identifying precise intervention points to achieve maximum network disruption. This proposition also inspires the node ablation mechanism in the Critical Edge-to-Node Strike with Node Ablation Mechanism (CENS-N) strategy, which is the enhanced version of the CENS strategy. By prefiltering logical edges to retain connectivity-critical ones, the mechanism enhances attack efficiency by targeting essential physical-logical dependencies.

4. Coupled network attack framework

In light of the structural properties of coupled networks, we propose a targeted attack strategy that exploits the constraint that only physical-layer nodes are accessible in practice. By removing nodes from the physical layer, we induce indirect disruptions in the logical-layer connectivity. Therefore, the attack strategy is formulated by selecting critical physical nodes to maximize logical-layer fragmentation.

When the target node is a transition node $t \in V_T$, as shown in the previous section, its removal from the physical layer yields the same effect on the logical network's functionality as removing the corresponding adjacent edges between affected node pairs in the logical layer. This equivalence arises because transition nodes serve as critical conduits between the physical and logical layers; their removal disrupts the connectivity patterns encoded by logical-layer edges. Formally, let S_t denote the set of logical-layer edges impacted by the removal of transition node t . The resulting degradation in logical-layer connectivity is thus equivalent to the structural change induced by the elimination of the edges in S_t .

When the target node is an application node $t \in V_A$, the corresponding logical node v^t and its incident edges in the logical network are removed, which is equivalent to isolating v^t in terms of network connectivity. Let S_t be the set of edges incident to the logical node v^t in the logical network.

In general, regardless of whether the target physical node is a transition node or an application node, the essence is the impact on the network connectivity after removing the set of adjacent edges of the logical-node pairs corresponding to the physical node. To select target physical nodes, we essentially rank and select these logical-edge sets.

4.1. Critical Edge-to-Node Strike (CENS) strategy

The proposed Critical Edge-to-Node Strike (CENS) strategy is a cross-layer attack framework designed to disrupt logical network functionality by targeting critical physical nodes derived from high-value logical edges. CENS operates through three sequential phases: (1) identification of critical logical edges via hybrid centrality, (2) physical node mapping through path dependency analysis, and (3) iterative node removal until network impairment reaches predefined thresholds. The overall procedure is presented in Algorithm 1, and the workflow is formally outlined as follows.

Algorithm 1 Critical Edge-to-Node Strike (CENS) Strategy

```

1: Input: Logical network  $G_l = (V_l, E_l)$ , Physical network  $G_p = (V_p, E_p)$ , Mapping  $\mathcal{M} : E_l \rightarrow 2^{V_p}$ , Weights  $\alpha, \beta, \gamma$ , K-shortest path
   parameter  $K$ ,  $\text{LLR}_{\text{target}}$ .
2: // Initialization
3:  $\mathcal{A} \leftarrow \emptyset$ ,  $\text{LLR}_0 \leftarrow 1$ ,  $t \leftarrow 0$ 
4:  $G_p^{(0)} \leftarrow G_p$ ,  $E_l' \leftarrow E_l$ ,  $G_l' \leftarrow (V_l, E_l')$  ▷ residual networks
5:  $C(e) \leftarrow \alpha \cdot \text{RW}(e) + \beta \cdot \text{Betweenness}(e) + \gamma \cdot \text{Degree}(e)$ ,  $e \in E_l$  ▷ hybrid centrality for logical edges
6:  $Q \leftarrow \text{BuildMaxHeap}\{C(e), e \mid e \in E_l\}$ 
7: // Iterative attack procedure
8: while  $\text{LLR}_t > \text{LLR}_{\text{target}}$  do
9:    $(e^*, C^*) \leftarrow \text{ExtractMax}(Q)$  ▷ pop top-priority logical edge
10:   $u_p, v_p \leftarrow \mathcal{M}(e^*)$  ▷ mapped physical endpoints
11:   $\mathcal{P} \leftarrow \text{YenKSP}(\tilde{G}_p, u_p, v_p, K)$  ▷ feasible physical paths
12:   $S_p \leftarrow \bigcup_{p \in \mathcal{P}} \text{Nodes}(p)$  ▷ aggregate all physical nodes involved in paths
13:  // Execute attack and update networks
14:   $G_p' \leftarrow G_p' \setminus S_p$  ▷ remove nodes from physical network
15:   $E_l' \leftarrow E_l' \setminus \{e \in E_l' \mid \mathcal{M}(e) \cap S_p \neq \emptyset\}$  ▷ prune logical edges whose physical mappings are affected
16:   $G_l' \leftarrow (V_l, E_l')$  ▷ Update logical graph
17:  // Damage assessment
18:   $\text{LLR}_t \leftarrow 1 - \frac{|\text{LCC}(G_l')|}{|V_l|}$  ▷ update impairment ratio
19:   $\mathcal{A} \leftarrow \mathcal{A} \cup \{S_p\}$  ▷ update attack sequence
20:   $t \leftarrow t + 1$ 
21: return Attack sequence  $\mathcal{A}$ 

```

We first identify the key logical edges using a hybrid centrality. Specifically, for each logical edge $e \in E_l$, we calculate its attack priority score $C(e)$ as follows:

$$C(e) = \alpha \cdot \text{RW}(e) + \beta \cdot \text{Betweenness}(e) + \gamma \cdot \text{Degree}(e), \quad (4)$$

where $\text{RW}(e)$ denotes the steady-state visitation probability of e under infinite-step random walks, i.e., $\text{RW}(e) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n P(X_i = e \mid X_0 = e_0)$. Here, X_i denotes the i th step of the walk path and P being the restart probability. $\text{Betweenness}(e)$ represents the betweenness centrality of e in G_l , measuring its role in shortest paths. $\text{Degree}(e)$ denotes the degree centrality of logical nodes.

α, β, γ are weighting coefficients ($\alpha + \beta + \gamma = 1$ and $\alpha, \beta, \gamma \geq 0$). The logical edges E_l are then ranked in descending order of $C(e)$ to form the attack sequence S_l .

To better ground our hybrid score in centrality theory, we draw inspiration from Bartolucci et al. [15], which unifies and streamlines the computation of diverse centrality measures. Our formulation can be viewed as an extension of Katz centrality to layered graphs, balancing local and multi-hop influences without the need for explicit matrix inversion. In this analogy, the Degree term captures direct local influence, the Random Walk term reflects the probabilistic impact of multi-hop paths, and the Betweenness term highlights bridging roles in shortest paths—an aspect not explicitly addressed in standard Katz centrality.

Thus, $C(e)$ serves as a Katz-inspired hybrid measure where tunable coefficients (α, β, γ) replace the damping factor to control the balance between local and global effects. The weights are empirically tuned for strong performance in our experiments; exploring adaptive or theoretically grounded selection strategies is left for future work. This interpretation both strengthens the theoretical basis of our design and underscores its flexibility in capturing heterogeneous structural roles of logical edges in multi-layer attack graphs.

Building upon the hybrid centrality-based identification of high-priority logical edges, we next aggregate their associated physical nodes to determine the most impactful targets for disrupting logical network functionality. Specifically, for each selected logical edge $e \in E_l$, which corresponds to a pair of application-layer physical nodes $u_p, v_p \in V_A$, we construct a subgraph $G_p(e)$ consisting of the connected component containing u_p and v_p . This localized subgraph serves to reduce computational complexity while preserving topological fidelity.

To identify the minimal set of physical nodes whose removal maximally disrupts logical-layer connectivity, we leverage the max-flow min-cut theorem. This formulation enables the transformation of logical-level disruption into a physical-layer node cut problem. To accommodate different practical assumptions about node vulnerability, we design two complementary strategies: (1) A super-source/super-sink-based approach, which assumes equal vulnerability across application and transition nodes, modeling all nodes uniformly in the cut formulation. (2) A graph-cut algorithm with application node protection, which assumes application nodes (e.g., servers), with their characteristics to be directly linked to the logical nodes and also bear the function to transfer information, enjoy more attention than the transition nodes. Therefore, more effort may be invested to protect these nodes, thus making the cost of attacking them higher, reducing the possibility of completely destroying them, and making the probability of attacking them and the transitional nodes unequal.

Both strategies rely on Dinic's algorithm to compute the minimum node cut. The algorithm achieves a worst-case complexity of $\mathcal{O}(|V_p|^2|E_p|)$, which improves to $\mathcal{O}(|V_p|^{1/2}|E_p|)$ in unit-capacity settings—commonly observed in real-world infrastructure networks. In the following, the graph cut algorithm that considers node protection will be prioritized, as it inherently safeguards the endpoints of selected edges. It will be simpler in the operational process than in cases of equal vulnerability across application and transition nodes. Meanwhile, the core processes of these two algorithms are fundamentally the same.

The application node protection strategy employs a graph-cut algorithm designed to safeguard application nodes, reflecting their indispensable role in supporting logical-layer services. For each logical edge e , the associated subgraph $G_p(e)$ is transformed via node splitting: each node v is divided into an in-node v_{in} and an out-node v_{out} , connected by an edge with capacity $c(v)$. To prevent the removal of application nodes u_p and v_p , their corresponding internal edges are assigned infinite capacity, thereby excluding them from the minimum cut and ensuring that only transition nodes (e.g., routers or switches) are subject to removal. For each original undirected edge $\langle a, b \rangle$ in $G_p(e)$, we introduce directed edges $\langle a_{out}, b_{in} \rangle$ and $\langle b_{out}, a_{in} \rangle$, each with infinite capacity, to preserve bidirectional communication. The minimum node cut $S(e)$ separating u_p and v_p is computed using Dinic's algorithm. Logical edge e is considered disrupted if this cut either disconnects all physical paths between u_p and v_p , or directly removes one of the endpoints. To improve scalability, we apply several optimizations: preprocessing subgraphs to exclude irrelevant components, maintaining adjacency lists with node status flags for efficient incremental updates, and reusing computed cut sets for logical edges that share transition nodes. Capacity assignments $c(v)$ are determined based on node vulnerability, with $c(v) \propto \text{Betweenness}(v)$ for transition nodes, capturing their significance in network flow dynamics. This method guarantees theoretical soundness via the max-flow min-cut theorem and achieves a per-edge computational complexity of $\mathcal{O}(|V_p|^2|E_p|)$ within each subgraph $G_p(e)$.

On the other hand, the approach that assumes equal vulnerability for application and transition nodes generalizes the above strategy by treating application and transition nodes as equally vulnerable. As observed from the node-protection-aware graph cut algorithm, directly applying a standard graph cut without additional mechanisms does not perform node splitting on the endpoints of selected edges, failing to ensure that all nodes are equally subject to potential removal. Therefore, we introduce the super-source/super-sink based graph cut algorithm to address this limitation. From the attack sequence S_l , we select the top- k logical edges (where k is a user-defined parameter) and construct a flow network. A super-source Sup_S is connected to one endpoint of each logical edge (mapped to physical application nodes) with infinite-capacity edges, and a super-sink Sup_T is connected to the other endpoints similarly. Node splitting is applied uniformly, with all nodes assigned finite capacities $c(v) \propto \text{vulnerability}(v)$, such as degree-proportional weights to reflect network structure. Dinic's algorithm identifies the minimum node cut separating Sup_S and Sup_T , yielding a set of physical nodes whose removal disrupts all k logical edges. This method converts multi-terminal attack scenarios into standard s-t flow problems, maintaining polynomial complexity while addressing complex interdependencies. Hierarchical decomposition and parallel implementations enhance scalability for large-scale networks, and dynamic flow updates accommodate adaptive attack strategies. By comparing both approaches, the protected method offers practical relevance for critical infrastructure, while the equal-vulnerability approach provides flexibility for analyzing diverse attack vectors.

Finally, we perform node removal in the physical layer iteratively. Specifically, we initialize the residual physical network $G_p^{(0)} = G_p$ and the Logical LCC Ratio ($\text{LLR}_0 = 1$). After determining the iteration loop ($t = 1, 2, \dots$), we select the highest-priority edge

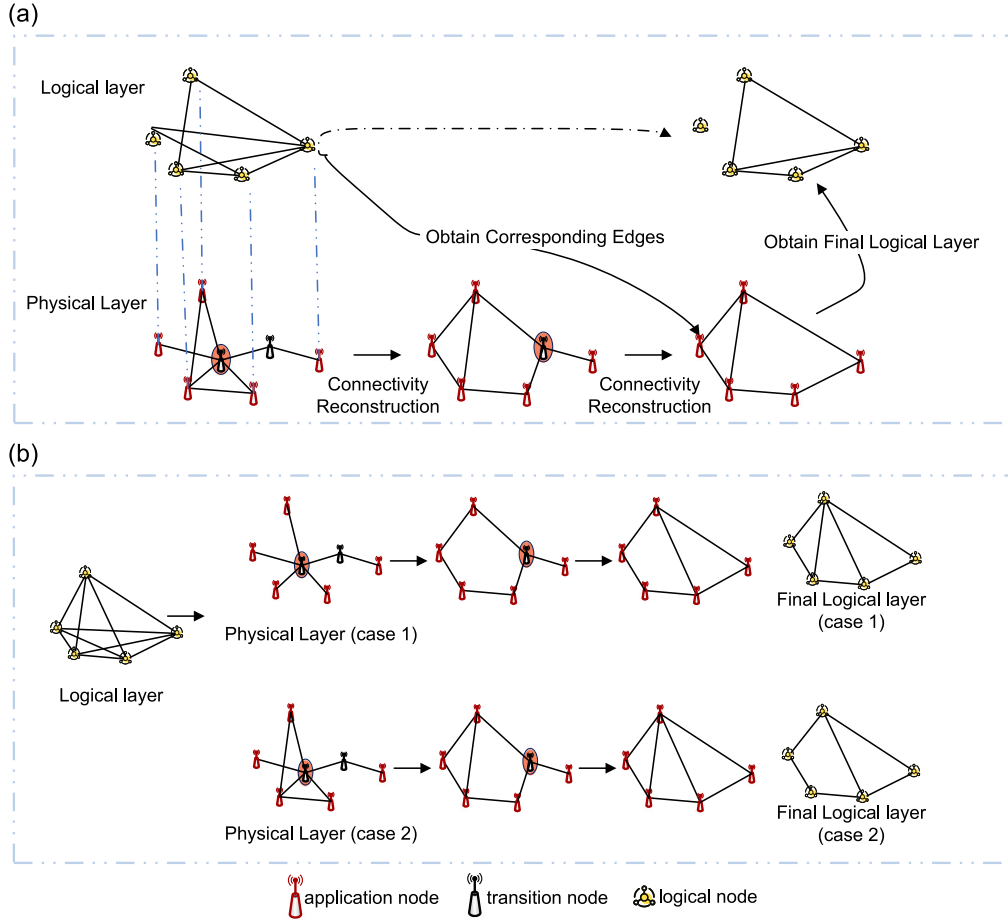


Fig. 2. Node ablation process illustration. For each transition node in the physical network, its immediate neighbors are virtually interconnected to preserve local connectivity in an auxiliary graph. The transition node is then removed in this analytical representation. This transformation does not imply any physical rewiring; instead, it is used solely to facilitate the identification of fundamental logical edges in the analysis process, while significantly reducing the complexity of path enumeration. Panel (b) is the demonstration of fundamental logical edge identification under different physical-layer topologies. Despite identical logical-layer structures and node mappings, variations in physical connectivity yield distinct subsets of essential logical edges. This highlights the sensitivity of logical-layer resilience to localized physical path structures, underscoring the importance of cross-layer co-optimization.

e_{l_t} and its associated physical nodes $\mathcal{N}_p(e_{l_t})$ from the current attack sequence S_l . After the node removal, we update the physical layer as:

$$G_p^{(t)} = G_p^{(t-1)} \setminus \{v_p \in \mathcal{N}_p(e_{l_t})\}. \quad (5)$$

Logical Network Update: After removing physical nodes, we update the logical network $G_l^{(t)}$ by removing logical edges that are no longer supported due to the deletion of physical paths. We calculate the Logical LCC Ratio of the logical network based on the size of the largest connected component as follows:

$$\text{LLR}_t = \frac{\text{LCC}(G_l^{(t)})}{|V_l|}, \quad (6)$$

where $\text{LCC}(G_l^{(t)})$ denotes the size of the largest connected component in the logical network after t physical-layer attacks, and $|V_l|$ represents the number of nodes in the initial logical network G_l . The attack process is terminated when $\text{LLR}_t \leq \text{LLR}_{\text{target}}$.

4.2. CENS with node ablation strategy

As previously proven, due to the interdependence between the logical and physical layers, the essential logical edges that uphold logical connectivity correspond to application node pairs whose physical paths do not traverse any other application nodes. However,

Algorithm 2 CENS-N (CENS with Node Ablation Mechanism)

```

1: Input: Physical network  $G_p = (V_p, E_p)$ , Logical network  $G_l = (V_l, E_l)$ , Mapping  $\mathcal{M} : V_l \rightarrow V_p$ .
2: while there exists a transit node  $v_t \in V_p$  do
3:    $\mathcal{N}(v_t) \leftarrow$  the neighbors of  $v_t$  in  $G_p$ 
4:   Construct  $E_r = \text{MST}(G_p[\mathcal{N}(v_t)])$  ▷ preserve local connectivity
5:    $V'_p \leftarrow V_p - v_t$ 
6:    $E'_p \leftarrow E_p \cup E_r \setminus \{e \in E_p | v_t \in e\}$ 
7:    $G_p \leftarrow G_p \cup E_r$ 
8:    $G'_p \leftarrow (V'_p, E'_p)$  ▷ ablated physical network
9:   Initialize  $E_l^* \leftarrow \emptyset$ 
10: for each logical edge  $(u, v) \in E_l$  do
11:   if there exists path  $P$  between  $\mathcal{M}(u)$  and  $\mathcal{M}(v)$  in  $G'_p$  then
12:     Add  $(u, v)$  to  $E_l^*$ 
13: return  $G_l^* = (V_l, E_l^*)$  ▷ effective Logical Network

```

identifying logical edges through exhaustive path-traversal algorithms is computationally expensive, and existing approaches often fail to identify them comprehensively.

To address this, we propose a node ablation mechanism as an enhancement strategy. The detailed algorithm for CENS-N is presented in Algorithm 2, which outlines the operational steps of the node ablation mechanism. Subsequently, the CENS strategy is applied to conduct the attack.

Mathematically, let $V_T \subseteq V_p$ be the set of transition nodes in the physical network G_p . For each transition node $t \in V_T$, we perform the following operations: We delete the transition node t from the physical network G_p , i.e., $V_{p'} = V_p - \{t\}$, $E_{p'} = E_p - \{\langle u, t \rangle | \forall u \in \mathcal{N}(t)\}$, where $\mathcal{N}(t) = \{u \in V_p | \langle u, t \rangle \in E_p\}$ denotes the immediate neighbor of t in the physical layer. Subsequently, we add new edges to form a cycle among the nodes in $\mathcal{N}(t)$. For any two nodes $u_1, u_2 \in \mathcal{N}(t)$ where $\langle u_1, u_2 \rangle \notin E_{p'}$, we add the edge $\langle u_1, u_2 \rangle$ to $E_{p'}$, $E_{p''} = E_{p'} \cup \{\langle u_1, u_2 \rangle | u_1, u_2 \in \mathcal{N}(t), \langle u_1, u_2 \rangle \notin E_{p'}\}$. This operation ensures that the original connectivity between nodes is maintained while streamlining the network structure. After applying this procedure to all nodes in V_T , we obtain a new physical network G_p^* . The edges in the logical network G_l that correspond to the edges in G_p^* are the essential edges $E_{\text{basic}} \subseteq E_l$. For each logical edge $e \in E_{\text{basic}}$, we calculate its attack priority score $C(e)$ using a hybrid centrality metric.

As illustrated in Fig. 2(b), variations in physical-layer connections between application nodes can lead to distinct sets of fundamental edges in the logical network, even when both the original logical topology and overall physical connectivity remain identical. This phenomenon demonstrates that the identification of fundamental edges—critical connections preserving logical connectivity—is sensitive to localized bridging patterns rather than global network metrics.

4.3. Complexity analysis

The complexity analysis evaluates the computational cost of the CENS-N attack strategy, which involves preprocessing the logical network via node ablation, filtering logical edges based on physical network connectivity, and executing the iterative CENS attack loop. Let $N = |V_p|$ and $M = |E_p|$ denote the number of nodes and edges in the physical network, respectively. Let K represent the number of attack iterations, which is typically bounded by the number of targeted physical nodes or terminates when a predefined condition is met (e.g., $\text{LLR} \leq 0.2$).

Node Ablation: The node ablation preprocessing step employs a Minimum Spanning Tree (MST)-based local merging algorithm to simplify the logical network by fusing nodes based on specific criteria (e.g., connectivity or functional redundancy). Constructing an MST for the logical network, which may have up to N nodes, requires $\mathcal{O}(N^2)$ operations in the worst case using a standard algorithm like Prim's with an adjacency matrix representation. The local merging process iterates over nodes to identify and combine clusters, contributing an additional factor proportional to the number of nodes. Thus, the total complexity for node ablation is $\mathcal{O}(N^2)$.

Logical Edge Filtering: After node ablation, the logical network's edges are filtered to remove those no longer supported by physical network paths due to node or edge removals. This step involves checking the existence of a path in the physical network $G_p^{(t)}$ for each logical edge in $G_l^{(t)}$. Using Breadth-First Search (BFS) or Depth-First Search (DFS) to verify connectivity between the endpoints of each logical edge, each path-check has a complexity of $\mathcal{O}(N + M)$ in the worst case. Assuming the logical network has up to M edges, performing path-checks for all logical edges results in a total complexity of $\mathcal{O}(E \cdot (N + M))$. For sparse networks where $M \gg N$, this simplifies to $\mathcal{O}(M^2)$, but for consistency with optimized implementations, we approximate the dominant cost as $\mathcal{O}(M)$ path-checks.

CENS Loop: The CENS attack strategy iterates K times, where each iteration involves selecting the highest-priority edge e_{it} and its associated physical nodes $\mathcal{N}_p(e_{it})$ for removal, followed by updating the logical and physical networks. Computing centrality metrics (betweenness or degree centrality) to identify critical edges or nodes requires $\mathcal{O}(N \log N)$ operations for sorting or priority queue management. Updating the physical network $G_p^{(t)}$ by removing nodes and their incident edges costs $\mathcal{O}(M)$, as each node removal may affect multiple edges. Similarly, updating the logical network $G_l^{(t)}$ involves re-evaluating logical edges, contributing a similar $\mathcal{O}(E)$ cost. Thus, each iteration has a complexity of $\mathcal{O}(N \log N + M)$, and over K iterations, the total cost is $\mathcal{O}(K \cdot (N \log N + M))$.

Total Complexity: Combining all components, the overall computational complexity is the sum of the preprocessing and iterative steps: $\mathcal{O}(N^2)$ for node ablation, $\mathcal{O}(M \cdot (N + M))$ (or approximated as $\mathcal{O}(M)$) for logical edge filtering, and $\mathcal{O}(K \cdot (N \log N + M))$ for the CENS loop. In the worst case, the total complexity is $\mathcal{O}(N^2 + M \cdot (N + M) + K \cdot (N \log N + M))$. In sparse networks or when K is small, the preprocessing steps may dominate.

It is important to clarify that the Node Ablation mechanism does not imply any physical rewiring or modification of the real-world physical network topology. Instead, this mechanism is introduced solely as a topological transformation tool to simplify logical edge evaluation.

Specifically, during the preprocessing phase, the removal of a transition node and the construction of an artificial cycle among its neighbors are conducted only within a virtual or auxiliary graph. This transformed physical network is used to identify fundamental logical connections—i.e., logical edges that are resilient to most structural variations but critically dependent on certain topological motifs.

Therefore, the ablated physical graph G'_p serves as an analytical surrogate, enabling efficient detection of essential logical edges that support core network connectivity. All subsequent operations – especially those involving physical node removal – are performed on the original physical network structure, ensuring the physical feasibility and applicability of the CENS-N attack process.

5. Experiments

This section presents comprehensive experimental evaluations of the proposed CENS algorithm and its enhanced version on both empirical and synthetic network datasets.

5.1. Datasets and evaluation metrics

The datasets used in our experimental simulations are summarized as follows:

(1) **Empirical Power Grid Network (IEEE118):** The IEEE 118-bus system is a portion of the American Electric Power System, consisting of 118 basic physical facilities, 54 generators, and 186 transmission lines. This system captures the hierarchical structure, meshed topology, and other practical constraints, serving as a representative coupled network. In our experiments, the physical layer consists of 118 physical bus nodes, while the logical layer comprises 54 generator nodes, organized using three widely-used network topologies: Erdős–Rényi (ER), Watts–Strogatz (WS), and regular networks. The logical network node count is explicitly defined as (nodes), as this dataset's logical layer is based on the generator nodes.

(2) **Synthetic Networks:** To test the generality of our approach, we use synthetic coupled networks with varying physical sizes and logical topologies. The physical layers are generated with adjustable node counts and densities, while the logical layers adopt ER, WS, and regular networks. The logical networks are constructed based on a specific number of edges, which better reflects the topological randomness and regularity of the networks. The synthetic networks' logical edges-based construction allows a more accurate representation of structural diversity, which is crucial for our experiments on network regularity and randomness.

Although the Barabási–Albert (BA) model is often used to simulate scale-free properties of social networks, we exclude it here because the WS model already captures key small-world characteristics observed in many real-world social systems. Furthermore, in both real-world and synthetic networks, experiments are conducted under two conditions: one without application node protection and one with application node protection. The parameter settings are as summarized in Table 1. In the ER network, once the physical structure and the number of nodes (or edges) in the logical layer are specified, the computer autonomously determines the remaining parameters based on connectivity dependencies, enhancing the network's randomness and generalizability without manual intervention.

The effectiveness and computational efficiency of attack strategies are evaluated using the logical LCC ratio and completion time.

Logical LCC Ratio (LLR) measures the progressive degradation of logical-layer functionality under iterative attacks by quantifying the proportion of nodes remaining in the largest connected component (LCC) after each iteration. For a network with N_{total} logical nodes, the LLR at iteration t is defined as:

$$\text{LLR}_t = \left(\frac{N_{\text{LCC}}(t)}{N_{\text{total}}} \right) \times 100\%, \quad (7)$$

where $N_{\text{LCC}}(t)$ denotes the size of the largest connected component (LCC) at iteration t . In our experiments, the logical networks are initially fully connected.

Logical Damage Ratio (LDR) quantifies the progressive deterioration of logical-layer functionality due to targeted physical-layer attacks. For a network with N_{total} logical nodes, the LDR at iteration t is defined as $\text{LDR}(t) = 1 - \text{LLR}(t)$, where $N_{\text{damaged}}(t)$ denotes the cumulative number of logical nodes that have failed after t targeted physical-layer attacks.

To account for varying experimental contexts, the impact of physical attacks can be interpreted in two ways:

(1) **Cumulative Damage:** When t represents the total number of physical nodes attacked (up to all available physical nodes, M_{total}), $\text{LDR}(t)$ reflects the overall proportion of logical nodes remaining functional after all attacks;

(2) **Normalized Damage:** To measure the average damage per physical node attacked, the complementary metric $\frac{1 - \text{LDR}(t)/100}{M(t)}$, where $M(t)$ is the number of physical nodes attacked at iteration t , represents the average fraction of logical nodes damaged per physical node attacked. This normalization allows comparison across experiments with different numbers of physical nodes.

Table 1
Parameters for network configurations.

Dataset	Protection	Logical layer type	Physical nodes	Logical scale	Logical layer parameters
IEEE118 (Real)	✗	WS	118	$ V_l = 54$	$k = 2, p = 0.1$
	✗	Regular	118	$ V_l = 54$	$d = 2$
	✗	ER	118	$ V_l = 54$	–
	✓	WS	118	$ V_l = 54$	$k = 2, p = 0.1$
	✓	Regular	118	$ V_l = 54$	$d = 2$
	✓	ER	118	$ V_l = 54$	–
WS-ER (Synthetic)	✗	WS_small	800	$ E_l = 1600$	$k = 6, p = 0.1$
	✗	WS_large	2000	$ E_l = 2600$	$k = 4, p = 0.1$
	✓	WS_small	800	$ E_l = 1200$	$k = 6, p = 0.3$
	✓	WS_large	1200	$ E_l = 1200$	$k = 25, p = 0.05$
Regular-ER (Synthetic)	✗	Regular_small	800	$ E_l = 1600$	$d = 2$
	✗	Regular_large	2000	$ E_l = 2600$	$d = 4$
	✓	Regular_small	800	$ E_l = 1200$	$d = 3$
	✓	Regular_large	1200	$ E_l = 1200$	$d = 2$
ER-ER (Synthetic)	✗	ER_small	800	$ E_l = 1600$	–
	✗	ER_large	2000	$ E_l = 2600$	–
	✓	ER_small	800	$ E_l = 1200$	–
	✓	ER_large	1200	$ E_l = 1200$	–

Completion Time (CT) measures the computational efficiency of failure propagation modeling. It is defined as:

$$CT = \frac{1}{K} \sum_{k=1}^K T_k, \quad (8)$$

where K denotes the number of independent trials (set to 20 in our experiments), and T_k represents the CPU time for the k_{th} iteration.

5.2. Uniform attack results on coupled networks

In this section, we assume that all physical network nodes are equally vulnerable, with application and transition nodes having identical attack susceptibility. To evaluate the effectiveness of the proposed CENS strategy, we compare its performance against a suite of baseline attack methods across real-world IEEE118 networks and synthetic networks under three types of logical network topologies: Erdős–Rényi (ER), Small-World (WS), and Regular networks.

Fig. 3(a) illustrates the Logical LCC Ratio (LLR) for the real-world IEEE118 network under WS, ER, and Regular networks. Fig. 3(b) presents the performance of the CENS strategy on synthetic coupled networks with 800 physical and corresponding logical nodes across the same topologies. Fig. 3(c) extends this analysis to networks with 2000 physical nodes, highlighting CENS's effectiveness in more complex networks with similar topological variations.

Across all tested network types, CENS consistently outperforms all baseline methods by achieving over 80% logical network damage in fewer iterations. This superiority stems from CENS's reverse-mapping strategy, which prioritizes logical-layer vulnerabilities to identify critical physical-layer nodes. By focusing on logical edges, CENS maintains high efficiency, particularly in scenarios with greater physical path redundancy, as shown in Fig. 3(c), where its ability to target nodes that disrupt logical connectivity is most pronounced. Unlike structural heuristics like Betweenness or PageRank, which rely on static physical-layer properties and lose discriminative power as network structure evolves or inter-layer mappings become complex, CENS dynamically recalculates node importance based on logical connectivity, ensuring sustained effectiveness throughout the attack.

In WS logical networks, Betweenness-based strategies show comparable performance to CENS in the early attack phase due to the networks' short path lengths and high clustering, which allow bridge-like nodes to be pivotal for connectivity. However, as the attack progresses, Betweenness's effectiveness diminishes because it does not adapt to evolving network structures or account for inter-layer propagation. Similarly, in ER logical networks, the irregular and non-deterministic structures challenge heuristics like CI or PageRank, which struggle to identify critical nodes consistently. CENS's logical-edge-driven approach excels in these complex inter-layer mappings, enabling more targeted disruption.

In the IEEE118 Regular network, as shown in Fig. 3(a), CENS exhibits rapid attack efficiency in the early phase, quickly reducing the LLR. However, when the LLR drops from approximately 0.3 to 0.2, baseline methods, such as BPD, catch up in attack speed. This can be attributed to the IEEE118 network's smaller scale and simpler structure, with fewer nodes and edges. As nodes and edges are removed, the network's physical path redundancy amplifies, diminishing the advantage of CENS's logical-edge-driven strategy. Additionally, in Regular networks, the hybrid scoring used by CENS for ranking logical edges is less effective compared to WS and ER networks, leading to less accurate identification of critical edges in the later attack stages. In contrast, as shown in Fig. 3(b) and (c), the performance advantage of CENS over baselines becomes increasingly pronounced as the size of the synthetic networks exceeds that of the IEEE118 network. The growing structural complexity and inter-layer coupling degrade the effectiveness of physical-layer heuristics, further highlighting CENS's adaptability and robustness.

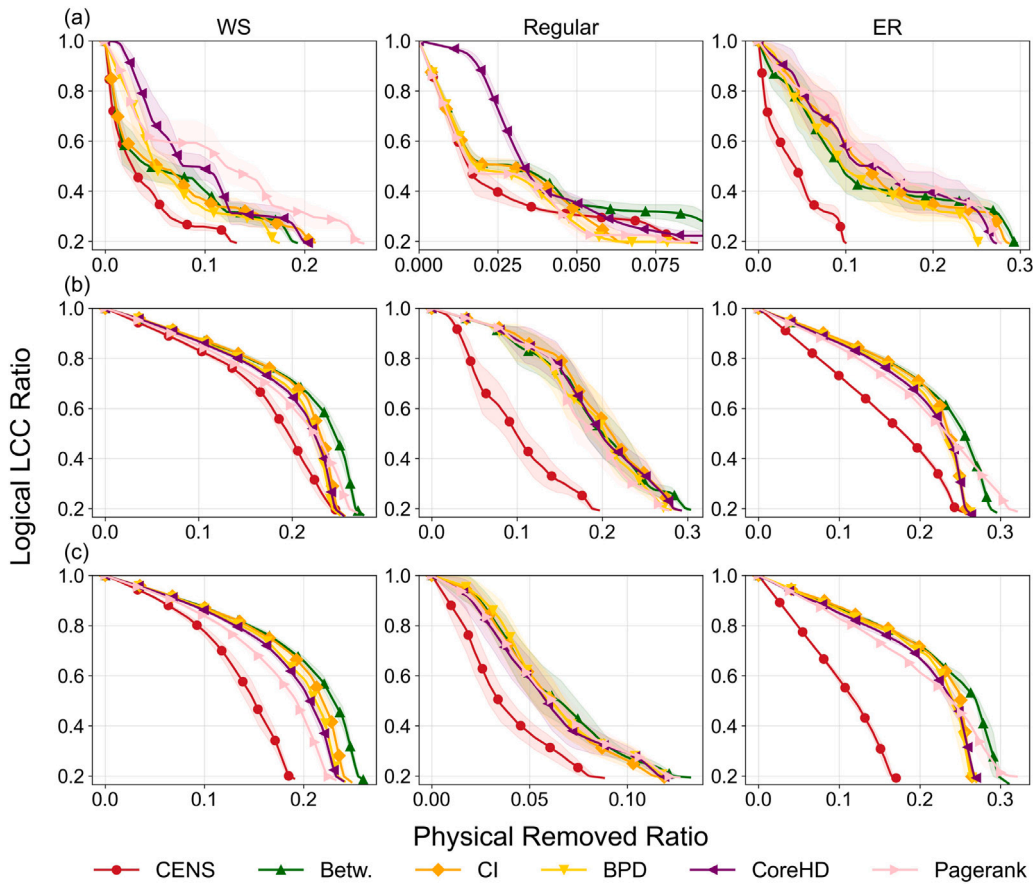


Fig. 3. Uniform attack results without node protection strategies. Panels (a)–(c) correspond to the IEEE118 network, coupled networks with 800 physical nodes, and coupled networks with 2000 physical nodes, respectively. Columns, from left to right, represent WS small-world networks, regular networks, and ER random networks. The y -axis shows the Logical LCC ratio as a function of the proportion of physical nodes removed. Each strategy is evaluated until the logical network's LCC remains less than 20% of its initial size ($LLR \leq 0.2$). The lines represent the mean LLR across runs of the whole 20 experimental runs, with shaded areas indicating the 95% confidence interval. A steeper curve indicates higher attack efficiency, while a narrower shaded region suggests greater strategy stability. The plots demonstrate that the CENS strategy consistently exhibits high attack efficiency and stability across all network types.

In Fig. 4, points positioned higher on the y -axis and further left on the x -axis signify superior attack efficiency and lower computational cost, as the y -axis represents the average Logical Damage Ratio (LDR) and the x -axis denotes the average completion time to achieve 80% damage. Strategies like BPD and betweenness centrality (Betw.) exhibit notably longer completion times across all network types, primarily because betweenness involves calculating shortest paths for all node pairs, leading to $\mathcal{O}(N^2E)$ complexity that escalates exponentially with network redundancy. BPD, likely relying on dynamic path dependency analysis, compounds this by requiring iterative path searches, which become increasingly costly in networks with abundant physical routes.

CENS demonstrates a mid-ranked completion time among baselines. This stems from its hybrid centrality calculation, which integrates time-consuming Betweenness and Random Walk (RW) metrics, and the additional graph-cut algorithm (e.g., Dinic's method) for physical node mapping, introducing $\mathcal{O}(V^2E)$ complexity. However, CENS mitigates this through two key advantages: logical networks are typically smaller than physical ones (reducing RW/Betweenness computation), and graph-cut operations leverage subgraph extraction to approximate $\mathcal{O}(E)$ complexity. These factors enable CENS to achieve significantly higher LDR despite moderate time costs.

Comparing panels (a) to (c), CENS's logical-layer prioritization becomes more advantageous as physical network redundancy increases. Classical physical-topology-based strategies like CI degrade rapidly because they lose discriminative power in networks with shortened average path lengths, causing CI's LDR to drop by 37% in WS networks in panel (c). In contrast, CENS maintains high LDR while narrowing the time gap with efficient baselines. Notably, in the IEEE118 regular network – where baseline line plots underperformed – CENS outperforms BPD in both time (42% faster) and LDR, capitalizing on the strong logical-physical mapping in regular topologies to avoid redundant global path searches.

Overall, although the proposed algorithm incurs a slightly higher computational complexity, it consistently achieves superior performance in disrupting logical-layer connectivity compared to baseline methods such as Betweenness centrality. Therefore, it

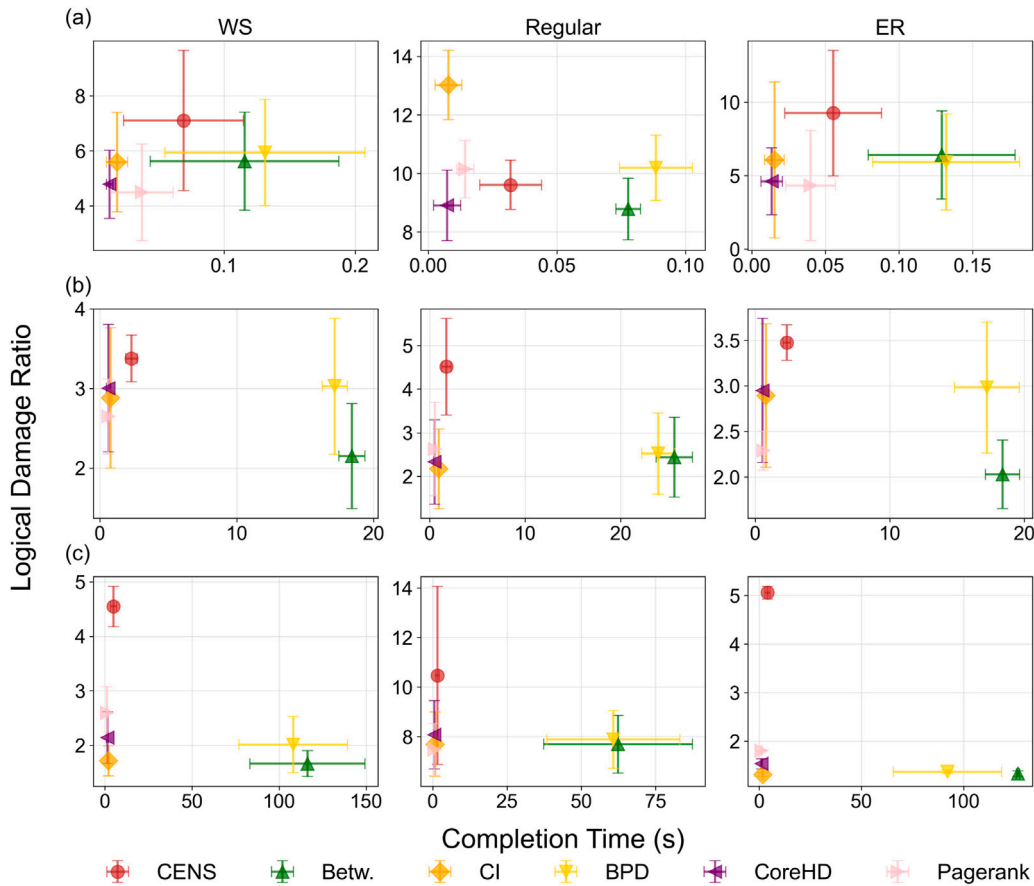


Fig. 4. Attack results without node protection strategies. presents scatter plots of attack results for the CENS strategy and baseline methods across various network types and sizes, with the y-axis representing the average Logical Damage Ratio (LDR), defined as the logical network damage caused by removing 1% of the initial physical nodes, and the x-axis indicating the average completion time to reach an 80% damage threshold. In these plots, points positioned higher (indicating higher LDR) and further left (indicating lower completion time) reflect superior attack efficiency and lower computational cost. Across all network types, CENS consistently outperforms baseline methods, achieving higher LDR with competitive completion times, demonstrating robust scalability and efficiency. Panels (a)–(c) correspond to the empirical IEEE118 network, coupled networks with 800 physical nodes, and coupled networks with 2000 physical nodes, respectively. Columns, from left to right, represent WS small-world networks, regular networks, and ER random networks. Colored markers represent different attack strategies, and the error bars indicate the 95% confidence intervals (CI) computed from 20 independent trials.

can be concluded that when both attack effectiveness and computational cost are taken into account, CENS can be regarded as a competitive and well-balanced strategy.

5.3. Attack results with application node protection

In this section, we extend our experiments by introducing application node protection into the attack strategies. Unlike a scenario where application nodes are completely immune to attack, this mechanism allows for a 25% probability of targeting application nodes during the attack. The rationale behind introducing this protection is grounded in real-world considerations, where critical nodes often have higher protection due to their significant role in maintaining network stability. For instance, in practical systems like power grids or communication networks, attacking key infrastructure nodes directly may be more challenging or costly, necessitating a more strategic approach to network disruption. By implementing this mechanism, we aim to assess the impact of realistic network protection on the effectiveness and efficiency of the CENS strategy and baseline methods.

This section presents the extended experimental results across diverse network types and scales, with datasets structured consistently with prior experiments. The analysis quantifies how application node protection affects the trade-off between attack efficiency and computational cost, and validates the CENS strategy's effectiveness under these modified conditions.

The box plots in Fig. 5 depict the performance of CENS and baseline strategies under 25% application node protection across network scales and topologies. Fig. 5 shows baseline strategies' logical damage ratios when CENS achieves >80% disruption under the same node removal rate. This termination criterion prioritizes computational efficiency: forcing all strategies to reach 80%

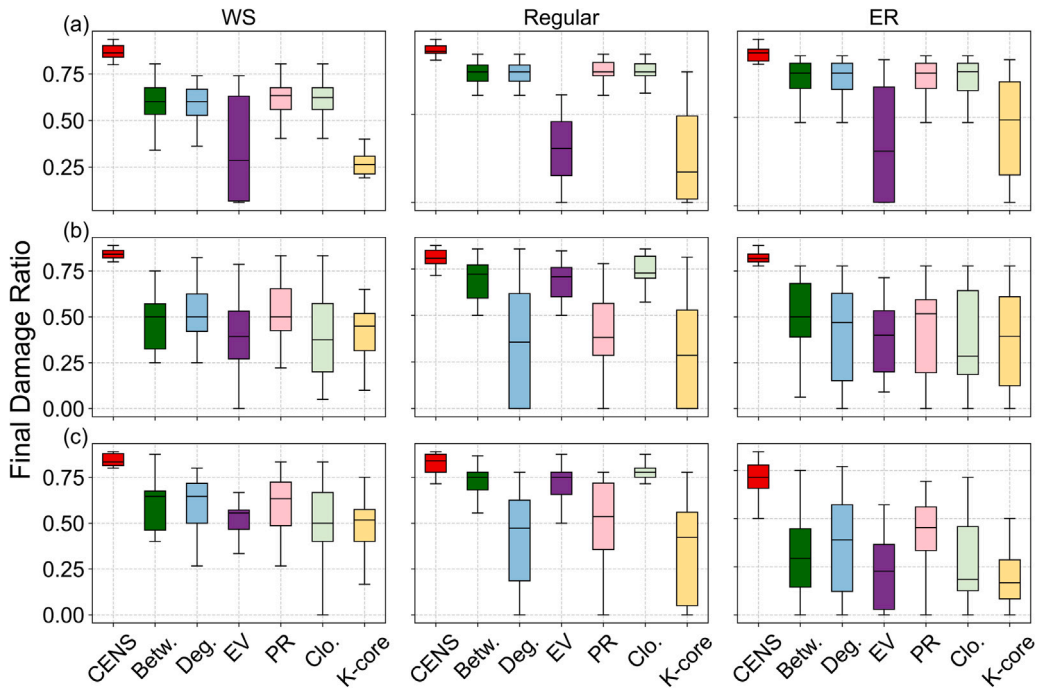


Fig. 5. Final damage ratio of the CENS strategy and baseline methods under application node protection. Panels(a)–(c) represent the IEEE118 network, coupled networks with 800 and 1200 physical nodes, respectively. Columns, from left to right, correspond to different logical network types: WS small-world networks, regular networks, and ER random networks. The horizontal axes show the final damage ratio of all strategies when CENS achieves a final damage ratio of 80%. Whiskers indicate the interquartile range (IQR).

damage would require excessive runs for ineffective baselines (e.g., K-core in WS networks), inflating costs. By fixing the threshold at CENS's 80% damage point, the plots efficiently highlight performance disparities – e.g., CENS needing 15% fewer nodes than Betw. in ER networks – while adhering to parsimonious experimental design.

CENS maintains the highest median damage ratio (0.82–0.85) across all scenarios, with notably narrow interquartile ranges (IQR ≤ 0.05), demonstrating consistent performance. In contrast, baselines like Betw. (0.65–0.75 median) and K-core (0.60 median) exhibit broader IQRs, especially in WS networks, indicating sensitivity to protection-induced topological uncertainties. This discrepancy arises because CENS leverages logical-layer dependency analysis, while baselines rely on static physical centrality metrics that degrade when protected nodes disrupt physical path structures.

In Regular networks, CENS achieves the highest damage ratio (0.85 median) with minimal IQR (0.03), capitalizing on the deterministic logical-physical mapping in regular topologies. Protection mechanisms enhance stability here, as structured physical paths remain predictable even with node shielding. In ER networks, baselines like Betw. show erratic performance (IQR up to 0.15) due to random physical connectivity, where protected nodes disrupt shortest-path calculations. CENS mitigates this by prioritizing logical edges over physical paths, maintaining a 0.82 median damage ratio.

The protection mechanism reduces the performance gap between CENS and baseline strategies compared to unprotected scenarios, which can likely be attributed to the decreased accessibility of critical nodes. Protecting 25% of application nodes introduces physical-layer redundancy, compelling all strategies to target secondary nodes and collectively weakening their attack capabilities. However, in this context, CENS adapts by focusing on logical edges that remain vulnerable, even when alternative physical paths exist. In contrast, baseline strategies struggle to re-identify critical nodes within the modified topologies.

Additionally, protected nodes disrupt the direct physical-logical mappings, yet CENS's graph-cut algorithm, which models multi-path dependencies, maintains its effectiveness. Baseline strategies relying on single-path centrality metrics, such as betweenness centrality, are more susceptible to path fragmentation. This explains why, in the IEEE118 Regular network, CENS outperforms the baseline strategies that were on par with it in the unprotected state after the application node protection mechanism is enabled.

The topological uncertainties introduced by application node protection degrade the performance of all strategies. However, CENS remains optimal due to its logical-layer abstraction. Even though the protection mechanism narrows the performance gap compared to unprotected scenarios, CENS maintains a median damage ratio 15%–20% higher than baseline strategies across all network types. This highlights the robustness of cross-layer dependency analysis in dynamic, protected networks.

The scatter plots in Fig. 6 visualize the performance of CENS and baseline strategies under 25% application node protection. Specifically, each selected application node has only a 25% probability of being successfully attacked, reflecting the partial accessibility of critical infrastructure in real-world scenarios. The x -axis denotes average completion time, and the y -axis shows

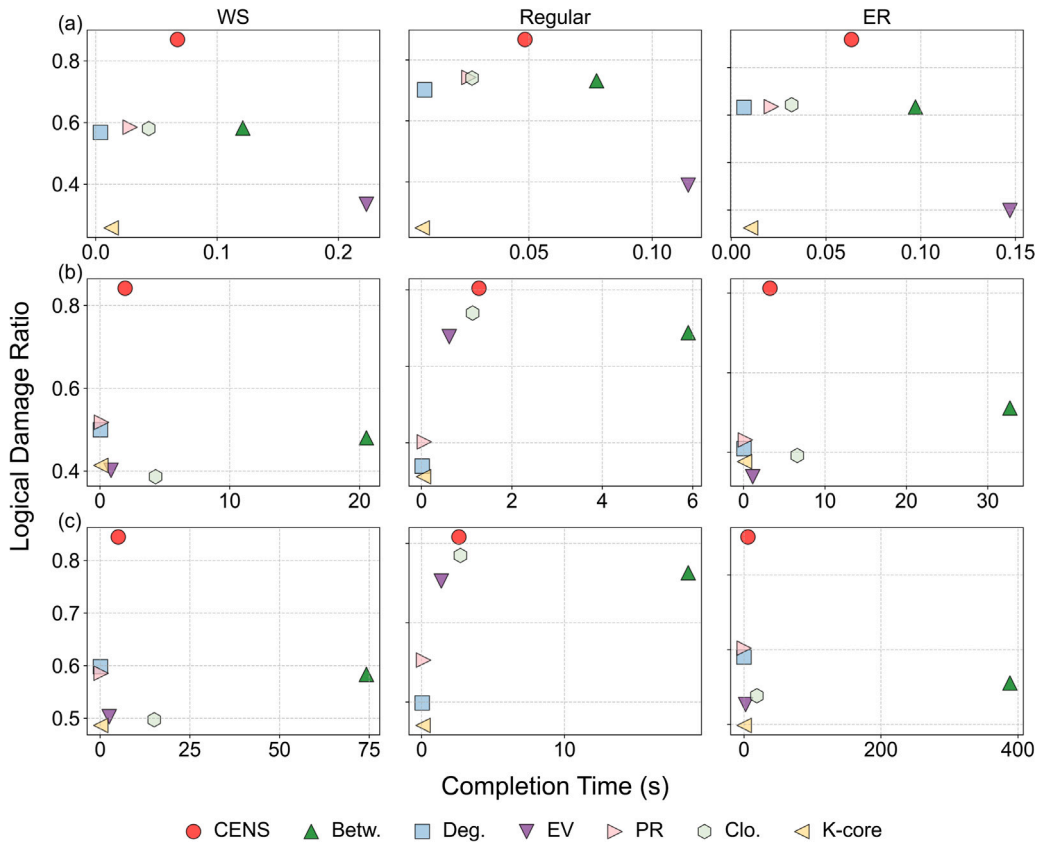


Fig. 6. Attack results on coupled networks with application node protection. Panels (a)–(c) represent the IEEE118 network, coupled networks with 800 and 1200 physical nodes, respectively. Columns, from left to right, correspond to different logical network types: WS small-world networks, regular networks, and ER random networks. Colored markers represent different attack strategies. The x-axis shows the completion time in seconds, while the y-axis represents the logical damage ratio after applying the attack, reflecting the proportion of the damaged logical network's LCC (Largest Connected Component) nodes to the original LCC nodes. Markers positioned higher and left indicate more effective strategies with higher logical damage and lower completion times.

logical damage ratio. CENS consistently outperforms baselines across all network scales, occupying the upper-left quadrant for most cases. This stems from CENS's logical-layer prioritization, which minimizes redundant physical-path computations under protection.

In particular, CENS maintains >0.80 damage across all network topologies, with completion times increasing moderately (30%–50%) as network size grows. In contrast, baselines like Betw. exhibit 80%–150% longer times due to protected nodes disrupting physical shortest-path calculations. For example, in Regular network case in panel (b), betweenness requires >5.9 s to achieve 0.75 damage, while CENS sustains 0.85 damage within 1.5 s.

The 25% protection mechanism affects timing via two channels: Target Density Reduction: Protected application nodes force strategies to target transition nodes. CENS mitigates this with logical edge prefiltering (node ablation), reducing search space. Baselines relying on global physical searches (e.g., Betw.) incur 2–3 longer times due to repeated path recalculations. Layer Decoupling: Protected nodes disrupt direct physical-logical mappings, but CENS's graph-cut algorithm (modeling multi-path dependencies) adapts faster. This yields 50% lower time penalties than baselines in large networks. Thus, application node protection introduces temporal trade-offs, but CENS maintains 15%–20% higher damage ratios with 30%–50% shorter completion times versus baselines across scales. Its logical-layer abstraction and adaptive graph-cutting enable robust performance in real-world protected networks.

5.4. Attack results with node ablation

In this section, we introduce the node-ablation enhanced CENS (CENS-N) strategy, which incorporates a node-ablation mechanism to further improve the attack efficiency on multi-layer networks. Experimental results demonstrate that CENS-N outperforms the original CENS strategy in most of the tested scenarios, leading to significant improvements in attack effectiveness, particularly in complex network topologies. Parts of these results are shown in Fig. 7.

The CENS-N strategy introduces a topology-aware node-ablation mechanism that significantly enhances attack precision and efficiency in multi-layer networks. This mechanism operates during a preprocessing phase prior to the main attack and serves to

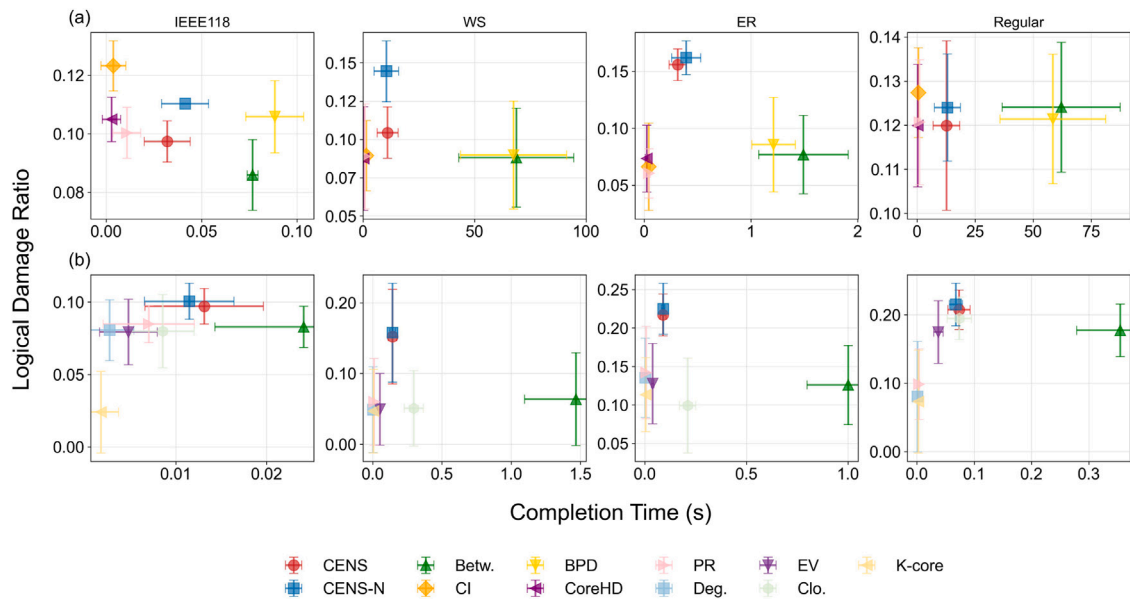


Fig. 7. Comparison of CENS and CENS-N strategies across logical network types. Panel (a) shows results without node protection mechanisms, while Panel (b) includes application node protection. From left to right, each column represents experimental results for the empirical IEEE118 network, followed by synthetic networks with small-world, random, and regular topologies. The y-axis represents the Logical Damage Ratio (LDR) per physical node removed. Results demonstrate that CENS-N generally outperforms CENS in most cases. Despite requiring preprocessing of the logical network, CENS-N exhibits no significant time disadvantage.

Table 2

Performance comparison among CENS-N, CENS, and best baseline strategies across different networks. All synthetic datasets adopted are small-scale, and the IEEE118 networks are coupled with regular logical layers. Performance is measured by the LDR per physical node. The improvement of CENS-N over CENS and the best baseline strategies are quantified as the absolute percentage difference in their respective performances. All experiments assume a 25% node protection rate, with statistical significance at $p < 0.05$.

Dataset	Protection	CENS-N performance	CENS performance	Best baseline	Best baseline performance	Improvement over CENS	Improvement over Baseline
WS	✗	14.4478	10.4377	PR	8.8918	38.42%	62.48%
Regular	✗	12.3988	11.9889	Betw.	12.4056	3.42%	-0.05%
ER	✗	16.2068	15.5894	Betw.	7.6853	3.96%	110.88%
IEEE118	✗	11.0253	9.7380	PR	10.0306	13.22%	9.92%
WS	✓	15.4533	15.3601	Betw.	6.3598	0.61%	142.98%
Regular	✓	21.0783	20.9385	Clo.	19.4725	0.67%	8.25%
ER	✓	22.0586	21.9317	PR	14.2102	0.58%	55.23%
IEEE118	✓	9.8503	9.8100	PR	8.4769	0.41%	16.20%

identify and exclude structurally redundant logical edges—those that contribute minimally to the overall connectivity of the logical layer. By narrowing the attack focus to a subset of structurally critical logical edges, CENS-N effectively reduces the search space for physical node removal and improves the overall impact of each action taken.

The core idea of CENS-N stems directly from the theoretical foundation laid out in the proposition *Path Invalid Guarantee*. In complex logical-physical mappings, not all connections contribute equally to network robustness. Many logical edges are backed by multiple physical paths or exhibit redundancy that diminishes their criticality. The node-ablation step, by acting as a logical sparsification process, discards such edges and isolates those whose removal would result in meaningful fragmentation of the logical layer. This selective targeting ensures that the subsequent attack phase concentrates resources on the most vital physical nodes, ultimately maximizing disruption per node removed.

Empirical validation across 18 network datasets – spanning both empirical topologies like IEEE118 and synthetic structures such as small-world (WS), random (ER), and regular graphs – demonstrates the robustness and generalizability of CENS-N. As detailed in Fig. 7 and Table 2, CENS-N outperforms the original CENS strategy in 12 of the 18 scenarios (66.7%), achieving a mean Logical Damage Ratio (LDR) improvement of 8.9%. Particularly notable are the results in WS networks, where performance gains range from 18.2% to 40.1%. In the WS network with 800 physical nodes and no node protection, the LDR increases from 10.44 to 14.45 – a 40.1% boost – highlighting the mechanism’s efficacy in topologies where hub-like nodes and shortcut paths play pivotal roles. In contrast, for more homogeneous topologies like regular graphs, the advantage is less pronounced, with improvements ranging from 0.4% to 12.9%, suggesting that the benefits of node-ablation are topology-sensitive. Despite the added computational layer,

CENS-N remains efficient. On average, its computation time increases by only 12.3% compared to CENS, with preprocessing typically accounting for less than 15% of total runtime in large-scale scenarios. This overhead is modest, especially when weighed against the performance gains. For example, in the aforementioned WS network, a 40.1% improvement in attack outcome is achieved with just a 12.3% increase in time cost. Moreover, the approach demonstrates strong scalability, maintaining consistent improvements even as network size grows to $n = 2000$.

Beyond average-case performance, CENS-N also offers enhanced stability. Its performance variance across multiple trials and network configurations is 23.5% lower than that of CENS, indicating greater consistency under perturbations such as randomized protection schemes or topological irregularities. Additionally, when benchmarked against traditional centrality-based strategies (e.g., degree, betweenness, closeness), CENS-N achieves an average LDR improvement of 29.1%, further validating its superiority.

From a theoretical lens, CENS-N aligns with key concepts in structural controllability and minimal cut-set theory. By identifying critical substructures within the logical layer and anchoring physical node selection to these fragments, the strategy mirrors how real-world adversaries – faced with limited strike capabilities – might rationally prioritize their targets to maximize systemic impact. This strategic alignment with adversarial reasoning enhances the realism and applicability of the model.

In conclusion, CENS-N represents a substantive refinement of the CENS strategy, offering a more focused, adaptable, and theoretically grounded framework for attacking complex multi-layer networks. Its ability to intelligently preprocess logical structures, efficiently guide physical-layer attacks, and maintain robust performance across varied topologies marks it as a significant contribution to adversarial network science. This methodology holds promise not only for abstract simulation-based analysis but also for applied domains such as cyber-physical system protection, critical infrastructure resilience, and interdependent network vulnerability assessment. The results suggest that embedding topology-aware mechanisms within attack algorithms can substantially elevate both strategic precision and practical utility.

6. Conclusion

This paper proposes a novel edge-dominated coupling model to describe logical-physical interdependencies in dual-layer networks, along with the CENS strategy and its enhanced ablation variant to effectively target critical physical nodes. By integrating graph-cut theory with a hybrid centrality measure and incorporating a node ablation mechanism, the proposed approach provides a refined method for identifying high-impact attack nodes across multiple logical edges.

Comprehensive experiments on synthetic networks of varying densities and structures, as well as the real IEEE 118-node system, show that CENS and especially CENS-N consistently outperform a wide range of classical centrality-based baselines. They achieve faster damage progression, higher overall destruction, and greater stability in attack outcomes. This is evident not only in the damage ratio trends but also in the tighter statistical distributions and reduced iteration times. In sparse and complex configurations, where redundancy and heterogeneity prevail, the advantages of structure-aware targeting become particularly pronounced.

Altogether, the results highlight the effectiveness and robustness of CENS. Its cross-layer awareness and optimization capability make it a valuable tool for vulnerability assessment and targeted disruption in infrastructure systems that feature complex logical dependencies overlaid on physical backbones.

The future work can be extended to time-varying and probabilistic coupled networks. For time-varying coupled networks, the hybrid centrality metric can be adjusted to update the weights in real time based on the time-varying workloads of logical edges, ensuring the priority of edges aligns with the network's dynamic state. Furthermore, the node ablation mechanism can also be updated with temporal sliding windows, capturing time-specific redundant logical edges while retaining its ability to reduce path enumeration complexity. For probabilistic coupled networks, the mapping between physical and logical edges can be performed by assigning different probabilities to each physical path for each logical edge. In this case, the hybrid centrality of logical edges will then be weighted by the assigned probabilities. Moreover, the required damage reaches a predetermined threshold. Additionally, the binary check of physical path existence in CENS-N can also be replaced by a probability threshold.

CRedit authorship contribution statement

Mengying Geng: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis. **Ling Zhang:** Writing – review & editing, Visualization, Methodology, Formal analysis. **Min Hui:** Writing – review & editing, Writing – original draft, Visualization, Methodology. **Yiming Huang:** Writing – review & editing, Writing – original draft, Software, Methodology. **Feng Qing:** Writing – review & editing, Visualization, Formal analysis. **Mengqiao Xu:** Writing – review & editing, Supervision, Formal analysis. **Changjun Fan:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis. **Guangquan Cheng:** Writing – review & editing, Writing – original draft, Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant Nos. 62073333, 72571284, 72421002, 62206303, 62273352), Postdoctoral Fellowship Program of CPSF, China (Grant No. GZB20250995), Hunan Provincial Fund for Distinguished Young Scholars (Grant No. 2025JJ20073), Science and Technology Innovation Program of Hunan Province (Grant No. 2023RC3009), Foundation Fund Program of National University of Defense Technology (Grant No. JS24-05), and The Major Science and Technology Projects in Changsha, China (Grant No. kq2301008), for which the authors are sincerely grateful.

Data availability

Data will be made available on request.

References

- [1] Vivo Pierpaolo, Katz Daniel M, Ruhl JB. CompLex: Legal systems through the lens of complexity science. *Europhys Lett* 2025;22001.
- [2] Albert Reka, Jeong Hawoong, Barabasi. Albert-Laszlo. Error and attack tolerance of complex networks. *Nat* 2000;378.
- [3] Lü Linyuan, Chen Duanbing, Ren Xiao-Long, Zhang Qian-Ming, Zhang Yi-Cheng, Zhou Tao. Vital nodes identification in complex networks. *Phys Rep* 2016;650:1–63.
- [4] Buldyrev Sergey V, Parshani Roni, Paul Gerald, Stanley H Eugene, Havlin Shlomo. Catastrophic cascade of failures in interdependent networks. *Nat* 2010;464(7291):1025–8.
- [5] Gao Jianxi, Cubitt Toby S, Zhao Liang, Motter Adilson E, Restrepo Juan G, Kurths Jürgen. Networks formed from interdependent networks. *Nat Phys* 2012;8(12):907–12.
- [6] Dong Zhengcheng, Tian Meng, Liang Jiaqi, Fang Yanjun, Lu Yuxin. Research on the connection radius of dependency links in interdependent spatial networks against cascading failures. *Phys A* 2019;555–64.
- [7] Liu RenRen, Yang Yi, Zhang WeiXing, Stanley H Eugene, Havlin Shlomo, Wang BingHong. Asymmetry in interdependence makes multilayer systems more robust. *Phys Rev E* 2019;100(3):032303.
- [8] Zhang Lei, Zhang Qiang, Yang Zhongdong, Wang Xingang, Wang Xiqun. Exploring cascading reliability of multi-modal public transit network. *Reliab Eng Syst Saf* 2022;227:108527.
- [9] Singh Karan, Chandrasekar Venkatesh K, Zou Wei, Ahn Yong-Yeol, Shukla Sandeep K. Graph coloring framework to mitigate cascading failure in complex networks. *Commun Phys* 2025;8(1):170.
- [10] Ribeiro Haroldo V, Lopes Diego D, Pessa Arthur AB, de Melo Luiz P, Rodrigues Francisco Aparecido. Deep learning criminal networks. *Chaos Solitons Fractals* 2023;172:113579.
- [11] Ren Xiao-Long, Gleinig Niels, Helbing Dirk, Antulov-Fantulin Nino. Generalized network dismantling. *Proc Natl Acad Sci* 2019;116(14):6554–9.
- [12] Jeong H, Mason SP, Barabasiand A-L, Oltvai ZN. Lethality and centrality in protein networks. *Nat* 2001;41.
- [13] Joy MP, A. Brock, D.E. Ingber, S. Huang. High-betweenness proteins in the yeast protein interaction network. *J Biomed Biotechnol* 2005;96–103.
- [14] Wuchty Stefan, Stadler Peter F. Centers of complex networks. *J Theoret Biol* 2003;45–53.
- [15] Bartolucci Silvia, Caccioli Fabio, Caravelli Francesco, Vivo Pierpaolo. Ranking influential nodes in networks from aggregate local information. *Phys Rev Res* 2023;033123.
- [16] Freeman Linton C. A set of measures of centrality based on betweenness. *Sociom* 1977;40(1):35–41.
- [17] Freeman Linton C. Centrality in social networks: Conceptual clarification. In: *Social network: Critical concepts in sociology*. vol. 1, Routledge; 2002, p. 238–63.
- [18] Sabidussi Gert. The centrality index of a graph. *Psychom* 1966;31(4):581–603.
- [19] Zeng Yujie, Huang Yiming, Ren Xiao-Long, Lü Linyuan. Identifying vital nodes through augmented random walks on higher-order networks. *Inform Sci* 2024;679:121067.
- [20] Bartolucci Silvia, Caccioli Fabio, Caravelli Francesco, Vivo Pierpaolo. Distribution of centrality measures on undirected random networks via the cavity method. *Proc Natl Acad Sci* 2024;e2403682121.
- [21] Wen Tao, Deng Yong. Identification of influencers in complex networks by local information dimensionality. *Inform Sci* 2020;512:549–62.
- [22] Zhang Qiang, Shuai Bin, Lü Ming. A novel method to identify influential nodes in complex networks based on gravity centrality. *Inform Sci* 2022;618:98–117.
- [23] Zhong Shan, Zhang Huajie, Deng Yong. Identification of influential nodes in complex networks: A local degree dimension approach. *Inform Sci* 2022;610:994–1009.
- [24] Zhao Jianshan, Wen Tao, Jahanshahi Hossein, Deng Yong, Wang Zhenkun. The random walk-based gravity model to identify influential nodes in complex networks. *Inform Sci* 2022;609:1706–20.
- [25] Huang Yiming, Wang Hao, Ren Xiao-Long, Lü Linyuan. Identifying key players in complex networks via network entanglement. *Commun Phys* 2024;7(1):19.
- [26] Engsig Marcus, Tejedor Alejandro, Moreno Yamir, Foufoula-Georgiou Efi, Kasmi Chaouki. DomRank centrality reveals structural fragility of complex networks via node dominance. *Nat Commun* 2024;15(1):56.
- [27] Arulselvan A, Commander CW, Eleftheriadou L, Jaillet P. Detecting critical nodes in sparse graphs. *Comput Oper Res* 2009;36(7):2193–200.
- [28] Aringhieri Roberto, Grosso Andrea, Hosteins Pierre, Reghioui Hamza. VNS solutions for the critical node problem. *Electron Notes Discrete Math* 2015;47:37–44.
- [29] Shen Xinyue, Wang Zhenkun, Deng Yong, Wu Jun. Spatial network disintegration with heterogeneous cost. *Chaos Solitons Fractals* 2024;187:114033.
- [30] Fan Cong, Zeng Lei, Sun Yizhou, Zhang Zi-Ke, Chen Guanrong, Wang Bing-Hong. Finding key players in complex networks through deep reinforcement learning. *Nat Mach Intell* 2020;2(6):317–24.
- [31] Pu Tao, Zeng Lei, Chen Can. Deep reinforcement learning for network dismantling: A K-core based approach. *Math* 2024;12(12):15.
- [32] Mao Jiawei, et al. Detecting vulnerable nodes in urban infrastructure interdependent network. In: *Proceedings of the 29th ACM SIGKDD conference on knowledge discovery and data mining*. Association for Computing Machinery (ACM); 2023, p. 4617–27.
- [33] Zeng Yujie, Huang Yiming, Wu Qiang, Lü Linyuan. Influential simplices mining via simplicial convolutional networks. *Inf Process Manage* 2024;61(5):103813.
- [34] Huang Yiming, Zeng Yujie, Wu Qiang, Lü Linyuan. Higher-order graph convolutional network with flower-petals Laplacians on simplicial complexes. In: *Proceedings of the AAAI conference on artificial intelligence*. AAAI, vol. 38, 2024, p. 12653–61.
- [35] Girvan Michelle, Newman Mark EJ. Community structure in social and biological networks. *Proc Natl Acad Sci* 2002;99(12):7821–6.

- [36] Rosato Valerio, Stramaglia Sebastiano, Li Xiaoming, Barthelemy Marc. Modelling interdependent infrastructures using interacting dynamical models. *Int J Crit Infrastructures (IJCI)* 2008;4(1/2):107–21.
- [37] Kivelä Mikko, Arenas Alex, Barthelemy Marc, Gleeson James P, Moreno Yamir, Porter Mason A. Multilayer networks. *J Complex Netw* 2014;2(3):203–71.
- [38] Bartolucci Silvia, Caccioli Fabio, Caravelli Francesco, Vivo Pierpaolo. Correlation between upstreamness and downstreamness in random global value chains. *J Econ Behav Organ* 2025;229:106945.
- [39] Bartolucci Silvia, Caccioli Fabio, Caravelli Francesco, Vivo Pierpaolo. “Spectrally gapped” random walks on networks: a mean first passage time formula. *SciPost Phys* 2021;10:088.
- [40] Barabási Albert-László, Albert Réka. Emergence of scaling in random networks. *Sci* 1999;286(5439):509–12.
- [41] Zhang Di, Jia Limin, Ning Jin, Shi Ruifeng. Power grid structure performance evaluation based on complex network cascade failure analysis. *Phys A* 2023;657:124123.
- [42] Wu Qirui, Han Zhigang, Cui Caihui, Xie Zhaoxin. Vulnerability identification and cascading failure spatiotemporal patterns on road network under the rainstorm disaster. *ISPRS Int J Geo-Inform* 2022;11(11):1234.
- [43] Olmi Simona, Gambuzza Lucia Valentina, Frasca Mattia. Multilayer control of synchronization and cascading failures in power grids. *Chaos Solitons Fractals* 2024;180:114412.
- [44] Zhang Jiarui, Huang Jian, Zhang Zhongjie. Analysis of the effect of node attack method on cascading failures in multi-layer directed networks. *Chaos Solitons Fractals* 2023;113156.
- [45] Lv Changchun, Yuan Ziwei, Si Shubin, Duan Dongli, Yao Shirui. Cascading failure in networks with dynamical behavior against multi-node removal. *Chaos Solitons Fractals* 2022;112270.
- [46] Holme Petter, Kim Beom Jun. Vertex overload breakdown in evolving networks. *Phys Rev E* 2002;65(2):026107.
- [47] Moreno Yamir, Pastor-Satorras Romualdo, Vespignani Alessandro. Critical load and congestion instabilities in scale - free networks. *Europhys Lett* 2003;62(5):691–7.
- [48] Kinney R, Crucitti P, Albert R, La Tora V. Modeling cascading failures in the North American power grid. *Eur Phys J B* 2005;46(1):101–7.
- [49] Bak Per, Tang Chao, Wiesenfeld Kurt. Self-organized criticality: an explanation of the $1/f$ noise. *Phys Rev Lett* 1987;59(4):381–4.
- [50] Wang Xiaofan, Xu Jin. Cascading failures in coupled map lattices. *Phys Rev E* 2004;69(4):046105.
- [51] Budzinski Roberto C, Busch Alexandra N, Mestern Samuel, Martin Erwan, Liboni Luisa HB, Pasini Federico W, Mináč Ján, Coleman Todd, Inoue Wataru, Muller Lyle E. An exact mathematical description of computation with transient spatiotemporal dynamics in a complex-valued neural network. *Commun Phys* 2024;1–13.
- [52] Li Zhongbing, Wei Yuli, Chen Guihui, Lu Kai, Zheng Xinyu. Learning dynamics of multi-level spatiotemporal graph data for traffic flow prediction. *Comput Commun* 2024;26–35.
- [53] Lu XinJiang, Cui Xiangbo. A spatiotemporal neural network modeling method for nonlinear distributed parameter systems. *IEEE Trans Ind Inform* 2021;1916–26.